NBTS 01B RM PHASE 2 NEET 2027

Q1.

Two trains each of length 150 m are moving in opposite direction at 36 km/hr and 72 km/hr respectively. Time taken by them to just cross each other completely is

(A) 4 s

(B) 6 s

(C) 8 s

(D) 10 s

Q2.

A particle has initial velocity $(2\hat{i} + 3\hat{j})$ and acceleration $(0.3\hat{i} + 0.2\hat{j})$. The velocity after 5 seconds will be

(A) $3\hat{i} + 4\hat{j}$

(B) $\frac{7}{2}\hat{i} + 4\hat{j}$

(C) $4\hat{i} + \hat{j}$

(D) $2\hat{i} + 3\hat{j}$

Q3.

Consider the following two statements

a. Among a given set of axis for a rigid body, the moment of inertia is always minimum about the one passing through the center of mass.

b. Perpendicular axis theorem for moment of inertia would be applicable for a solid sphere.

The correct statement(s) is/are

(A) Only a

(B) Only b

(C) Both a and b

(D) Neither a nor b

Q4.

A particle is thrown from a height h with speed u. The particle will hit the ground with maximum speed if

(A) It is thrown vertically downward

(B) It is thrown horizontally

(C) It is thrown vertically upward

(D) It is thrown in any direction

Q5.

A person sitting firmly over a rotating stool has his arms folded. If the person stretches his arms, the angular momentum about axis of rotation

(A) Increases

(B) Decreases

(C) Remains unchanged

(D) Become zero suddenly

Q6.

A ball is moving with velocity 5 m/s towards a heavy wall moving towards the ball with speed 2 m/s. Assuming collision to be elastic, what will be the speed of ball immediately after the collision?

(A) 9 m/s

(B) 3 m/s

(C) 2.5 m/s

(D) 5 m/s

Q7.

Two solid spheres *A* and *B* of radii *r* and 2*r* of same material are kept in contact, then the distance of centre of mass from the centre of sphere *A* will be

(A) $\frac{8r}{3}$

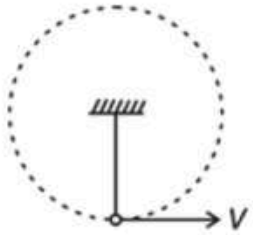
(B) $\frac{4r}{3}$

(C) $\frac{5r}{2}$

(D) $\frac{7r}{2}$

Q8.

A particle of mass 0.5 kg is rotating in a vertical circular path with the help of an inextensible string of length 1 m as shown in the figure.



If value of speed of particle at lowest point of circular path is 8 ms^{-1} then tension in the string at this instant will be (g = 10 m/s^2)

(A) 30 N

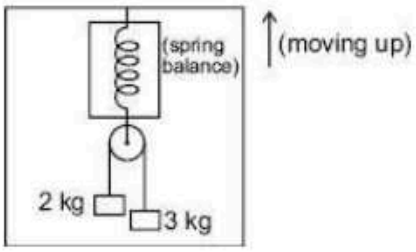
(B) 48 N

(C) 37 N

(D) 32 N

Q9.

A spring balance is attached to roof of an elevator moving upward with a retardation of 10 m/s^2 as shown in the figure. The reading of the spring balance is (g = 10 m/s^2)



(A) 5 kg weight

(B) Zero

(C) 20 kg weight

(D) 10 kg weight

Q10.

The moment of inertia of a uniform cylinder about an axis is given by $\frac{3}{2}MR^2$. The value of radius of gyration of the cylinder about this axis will be (*M* and *R* are mass and radius of cylinder)

(A) $\frac{\sqrt{3}}{2}R$

(B) $\sqrt{\frac{2}{3}}R$

(C) $\frac{\sqrt{3}}{\sqrt{2}}R$

(D) $\frac{R}{\sqrt{3}}$

Q11.

A particle initially at origin and moves in *x*-*y* plane with constant acceleration α in y-direction. Its equation of motion is $y = 2\beta x^2$. The velocity component of particle in x-direction is (where β is constant)

(A) $\sqrt{\frac{\alpha}{\beta}}$

(B) $\sqrt{\frac{\alpha}{2\beta}}$

(C) $\sqrt{\frac{2\alpha}{\beta}}$

(D) $\frac{1}{2}\sqrt{\frac{\alpha}{\beta}}$

Q12.

The concept of inertia is explained in

(A) Newton's first law

(B) Newton's second law

(C) Newton's third law

(D) All of these

Q13.

The mass of a cylindrical body depend on density d of material, its diameter D and length L as $M \propto d^x D^y L^z$. If maximum percentage error in measurement of density, diameter and length are $m\%$ $n\%$ and $l\%$ respectively, then maximum percentage error in calculation of mass is

(A) $(m + l + n)\%$

(B) $(mx + yn + lz)\%$

(C) $\left(\frac{m}{x} + \frac{n}{y} + \frac{l}{z}\right)\%$

(D) $(mx + ly + nz)\%$

Q14.

The equation of trajectory of a projectile is given as $y = 2x - \frac{x^2}{2}$. The maximum height of projectile is (Symbols have usual meanings and SI unit)

(A) 4 m

(B) 1 m

(C) 2 m

(D) 10 m

Q15.

If $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + \hat{j} + 7\hat{k}$ then component of $\vec{a}$ along $\vec{b}$ has magnitude

(A) $\frac{33}{\sqrt{47}}$

(B) $\frac{23}{\sqrt{47}}$

(C) $\frac{23}{\sqrt{59}}$

(D) $\frac{33}{\sqrt{59}}$

Q16.

A torque $\tau = 6t$ acts on a body, where τ is in N m and t is in seconds. The total angular impulse acting on the body in time interval $t = 0$ to $t = 2$ s is

(A) 22 kg m^2/s

(B) 2 kg m^2/s

(C) 12 kg m^2/s

(D) 10 kg m^2/s

Q17.

A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. If α and β are constants, then dimensions of [αβ] is:

(A) $[M^0L^2T^{-3}]$

(B) $[M^2L^2T^{-3}]$

(C) $[M^2L^2T^{-7}]$

(D) $[ML^2T^{-2}]$

Q18.

The position of a particle is given by $\vec{r} = \frac{3}{2}t^2\hat{i} + 2t^2\hat{j} + 5\hat{k}$ m, where t is time in seconds. The acceleration of the particle at t = 1 s is

(A) $3\ m\ s^{-2}$

(B) $4\ m\ s^{-2}$

(C) $5\sqrt{2}\ m\ s^{-2}$

(D) $5\ m\ s^{-2}$

Q19.

If force F, energy E and velocity v are chosen as fundamental quantities, the dimensional formula for acceleration will be

(A) $[FvE]$

(B) $[Fv^2E^{-1}]$

(C) $[FvE^{-1}]$

(D) $[F^{-1}v^{-2}E^{1}]$

Q20.

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : The average and instantaneous velocities have same value in a uniform motion.

Reason (R) : In uniform motion, the velocity of an object increases uniformly.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both **A** and **R** are true and **R** is the correct explanation of **A**

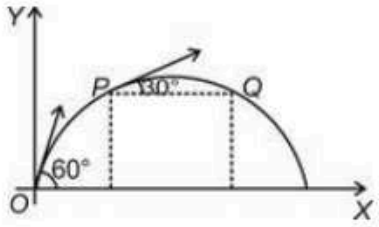
(B) Both **A** and **R** are true but **R** is **not** the correct explanation of **A**

(C) Both **A** and **R** are false

(D) **A** is true but **R** is false

Q21.

A particle is projected from ground with initial velocity $18\sqrt{3}\ \mathrm{m/s}$ making an angle 60° with the horizontal *x*-axis as shown in figure. The time taken by particle in moving from *P* to *Q* (on the same horizontal level) is (take $g = 10\ \mathrm{ms}^{-2}$)



(A) 1.8 s

(B) 3.6 s

(C) 5.4 s

(D) 7.2 s

Q22.

Which of the conversion of units among the following is correct?

(A) 1 hp = 740 W

(B) 1 erg = 10^{-5} J

(C) 1 kWh = 3.6×10^{6} J

(D) 1 cal = 3.186 J

Q23.

If a projectile of mass *m* is projected (from ground) with velocity *u* at angle θ with horizontal, then

A. Torque of weight about point of projection, when it is striking the ground, is zero.

B. Work done by gravity in entire time of flight is zero.

C. Angular momentum of projectile about point of projection, when it is striking the ground, is zero.

The correct statement(s) is/are

(A) A only

(B) B only

(C) A and C only

(D) A, B and C

Q24.

A machine gun fires 10 bullets per second with speed 10 m/s. If mass of each bullet is 300 g, then the force required to keep the gun stationary is

(A) 40 N

(B) 10 N

(C) 20 N

(D) 30 N

Q25.

A man can swim at the rate of 2.5 km/h in still water. A river of width 1 km flowing at the rate of 1.5 km/h. At what angle he should jump with the flow of river so that he reaches to the exactly opposite point on the bank?

(A) 37°

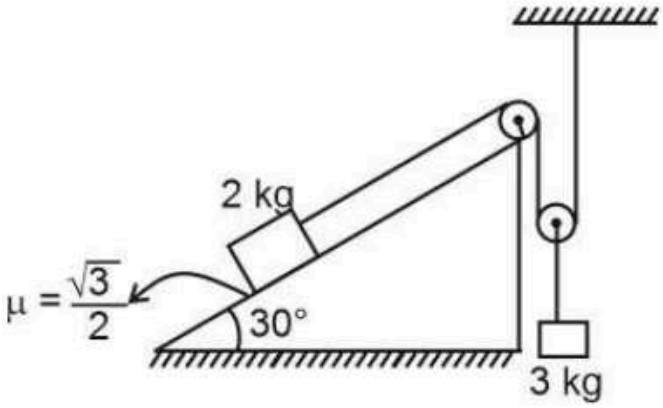
(B) 127°

(C) 53°

(D) 70°

Q26.

Consider an arrangement of ideal string and pulleys and a block kept on an inclined plane. If the system is released from rest then match the entries in Column-I with correct entries in Column-II. (g = 10 m/s^2)



	Column - I	**Column - II** (All quantities in SI unit)
A	Acceleration of 2 kg mass (m/s^2)	P 5
B	Acceleration of 3 kg mass (m/s^2)	Q Zero
C	Tension in string connecting 3 kg mass (N)	R 2
D	Frictional force on 2 kg mass (N)	S 30

(A) A → Q, B → Q, C → S, D → P

(B) A → Q, B → Q, C → P, D → S

(C) A → R, B → P, C → Q, D → S

(D) A → P, B → R, C → S, D → Q

Q27.

A particle starts from origin, moves along *x*-axis to the point (5 m, 0) and then returns along the same line to the point (–10 m, 0). The

displacement of particle during the trip is

(A) 10 m

(B) –10 m

(C) 15 m

(D) –20 m

Q28.

If Planck's constant h is taken into a new system of units in which new unit of mass is 10 g, new unit of length is 5 m, new unit of time is 100 s, then the value of Planck's constant in new system is
(Given $h = 6.6 \times 10^{-34}$ Js)

(A) 6.6×10^{-34} new unit

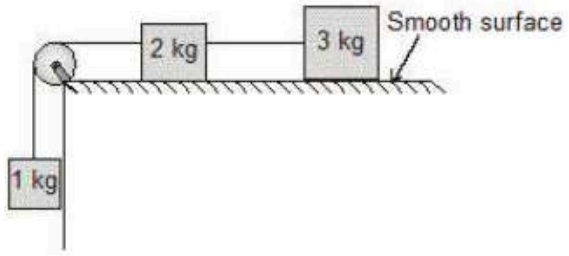
(B) 2.64×10^{-31} new unit

(C) 26.4×10^{-31} new unit

(D) 26.4×10^{-33} new unit

Q29.

In ideal mass pulley system, three masses are connected as shown in figure. Acceleration of the blocks is [g = acceleration due to gravity]



(A) $\frac{g}{4}$

(B) $\frac{g}{6}$

(C) $\frac{g}{2}$

(D) $\frac{g}{3}$

Q30.

The number of significant figures in the measured value 0.4010 g is

(A) One

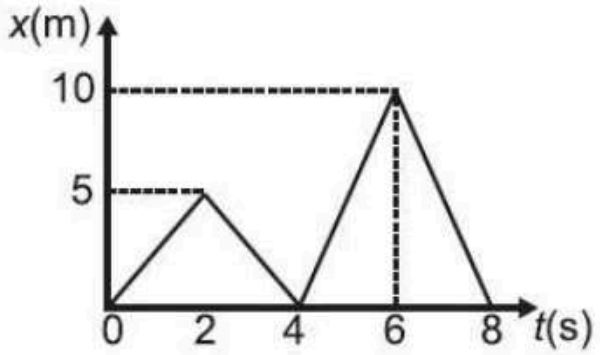
(B) Two

(C) Four

(D) Three

Q31.

The position-time (*x*-*t*) graph for a particle moving along x-axis is as shown in the graph. Average speed of the particle between time $t = 0$ and $t = 8$ s is



(A) Zero

(B) 8 m/s

(C) 3.75 m/s

(D) 4.25 m/s

Q32.

In multiplication or division of measured values, the final result should retain as many significant figures as are there in

(A) Original number with maximum decimal places

(B) Original number with the maximum significant figures

(C) Original number with the least significant figures

(D) Original number with least decimal places

Q33.

In the one-dimensional motion of a particle, the relation between position x and time t is given by $x^2 = t - 8x$. The retardation of particle is

(Consider SI units for all.)

(A) $\frac{1}{4(x+1)^3}$

(B) $\frac{1}{4(x+4)^3}$

(C) $\frac{1}{8(x+4)^3}$

(D) $\frac{1}{8(x-1)^3}$

Q34.

If an object is moving with constant speed, then its acceleration

(A) Must be zero

(B) May be uniform

(C) May be variable

(D) Must be variable

Q35.

Which among the following options is not correct about torque?

(A) Torque is a vector quantity

(B) Torque of a force $\vec{F}$ acting on a point having position vector $\vec{r}$ is given by $\vec{\tau} = \vec{r} \times \vec{F}$

(C) If a body is rotating about a fixed axis there must be a

(D) Dimensional formula of torque is same as that of work

non-zero torque acting on the body

Q36.

Consider the following statements and choose the correct option.

Statement A: Static friction is always greater than sliding friction.

Statement B: On a smooth horizontal floor, a vehicle can't take a turn.

(A) Both statement A and statement B are false

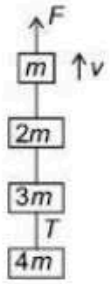
(B) Statement A is true but Statement B is false

(C) Both statement A and statement B are true

(D) Statement A is false but Statement B is true

Q37.

Four blocks with masses m, $2m$, $3m$ and $4m$ are connected by light string as shown in the figure. After an upward force F is applied on block of m, the masses move upward at constant speed v. The tension (T) in the string between $3m$ and $4m$ is



(A) $4\ mg$

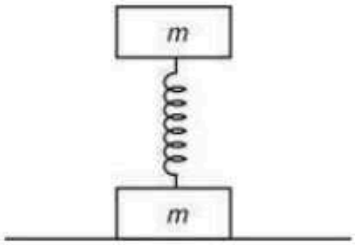
(B) $7\ mg$

(C) $2\ mg$

(D) $3\ mg$

Q38.

A system consists of two identical blocks, each of mass m connected by a massless spring of force constant k. Initially the system is in equilibrium as shown in the figure. The further compression that must be provided to the spring such that the lower block just lifts off the ground is



(A) $\frac{mg}{k}$

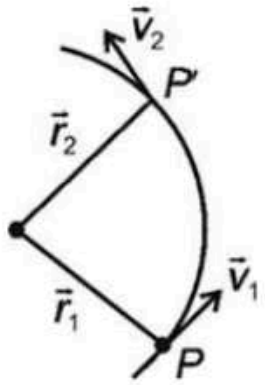
(B) $\frac{3mg}{k}$

(C) $\frac{4mg}{k}$

(D) $\frac{2mg}{k}$

Q39.

A particle moving with constant speed v along a circle of radius r covers distance from P to P' as shown.



If $\vec{\Delta r} = \vec{r_2} - \vec{r_1}$ and $\vec{\Delta v} = \vec{v_2} - \vec{v_1}$, then

(A) $\vec{\Delta v} = \frac{v}{r}\vec{\Delta r}$

(B) $\vec{\Delta v} = \frac{r}{v}\vec{\Delta r}$

(C) $\vec{\Delta v} = \vec{\Delta r}$

(D) $\vec{\Delta v} = r\vec{\Delta r}$

Q40.

Position of a particle moving along *x*-axis is given as $x = 8t^2$. The graph between velocity (v) and time (t) for the particle will be

(A) Ellipse

(B) Parabola

(C) Hyperbola

(D) Straight line

Q41.

The angular position of a point over a rotating flywheel is changing according to the relation, $\theta = (2t^3 - 3t^2 - 4t - 5)$ radian. The angular acceleration of the flywheel at time, $t = 1$ s is

(A) Zero

(B) 4 rad/s^2

(C) 6 rad/s^2

(D) 12 rad/s^2

Q42.

Consider the following statements:

Statement (A): The total linear momentum is conserved at each instant of any collision.

Statement (B): The kinetic energy of the system is conserved at each instant of any collision.

Choose the **correct** option.

(A) Only statement (A) is correct

(B) Only statement (B) is correct

(C) Both statements are correct.

(D) Both statements are incorrect

Q43.

Which of the following equation is dimensionally incorrect? (Symbols have their usual meaning)

(A) $T = 2\pi\sqrt{\frac{l}{g}}$

(B) $v^2 = u^2 + 2aS$

(C) $S = ut + \frac{1}{2}at$

(D) $v = u + at$

Q44.

Under the action of a force, a 1 kg body moves such that its position x as a function of time is given by $x = \frac{t^4}{4}$, (where x is in meter and t is in seconds). The work done by force in first two seconds is

(A) 8 J

(B) 16 J

(C) 32 J

(D) 64 J

Q45.

If $k\hat{i} + \frac{1}{2}\hat{j}$ is a unit vector, then k can be equal to

(A) $\frac{\sqrt{3}}{2}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

Q46.

The correct order of bond angles in the following compounds/species is :

(A) $CO_2 < NH_3 < H_2O < \overset{+}{N}H_4$

(B) $H_2O < NH_3 < \overset{+}{N}H_4 < CO_2$

(C) $H_2O < \overset{+}{N}H_4 < NH_3 < CO_2$

(D) $H_2O < \overset{+}{N}H_4 = NH_3 < CO_2$

Q47.

The bond order of P – O bond in PO_4^{3-} is

(A) 1.5

(B) 2

(C) 1.25

(D) 1.75

Q48.

Match the List-I with List-II.

List I	List II
(a) Nickel	(i) Non-metal
(b) Chlorine	(ii) Lanthanoid
(c) Germanium	(iii) Transition metal
(d) Samarium	(iv) Metalloid

Choose the correct match.

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(C) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(D) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

Q49.

Consider the following statements about photoelectric effect

(a) There is no time lag between the striking of light beam and the ejection of electrons from metal surface.

(b) The number of electrons ejected is directly proportional to the frequency of light.

(c) The kinetic energy of ejected electrons increases with the increase of intensity of light used.

The correct statement(s) is/are

(A) (a) only

(B) (a) and (b) only

(C) (b) and (c) only

(D) (a), (b) and (c)

Q50.

Match the species given in List I with their bond order given in List II.

List I	List II
(a) O_2	(i) 2.5
(b) O_2^+	(ii) 2
(c) O_2^-	(iii) 3
(d) O_2^{2+}	(iv) 1.5

Choose the **correct** option.

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(C) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(D) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

Q51.

If the radius of Bohr orbit for second excited state of Li^{2+} ion is x Å then radius (in Å) of Be^{3+} ion in its ground state will be

(A) $\frac{x}{2}$

(B) $\frac{4x}{3}$

(C) $\frac{x}{12}$

(D) $\frac{x}{3}$

Q52.

Match prefixes given in List-I with multiple given in List-II

List-I	List-II
a. mega	(i) 10^{18}
b. giga	(ii) 10^{6}
c. exa	(iii) 10^{12}
d. tera	(iv) 10^{9}

The correct match is

(A) a(iii), b(ii), c(i), d(iv)

(B) a(ii), b(iv), c(iii), d(i)

(C) a(iv), b(ii), c(i), d(iii)

(D) a(ii), b(iv), c(i), d(iii)

Q53.

28 g of $N_2(g)$ and 3 g of $H_2(g)$ are mixed to produce $NH_3(g)$ according to the following reaction

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

The correct option regarding the above reaction is

(A) 22.4 L of $NH_3(g)$ is formed at STP

(B) 34 g $NH_3(g)$ is formed

(C) 0.7 moles of $N_2(g)$ remain unreacted

(D) 0.25 moles of $H_2(g)$ remain unreacted

Q54.

Consider the following statements about PCl_5.

(a) It has sp^3d hybridisation.

(b) Equatorial bonds are shorter than axial bonds.

(c) Maximum number of atoms lying in one plane is 5.

The **correct** statements are

(A) (a) and (b) only

(B) (b) and (c) only

(C) (a) and (c) only

(D) (a), (b) and (c)

Q55.

The frequency of electromagnetic radiation of wave number 10^4 cm^{-1} is

(A) 2×10^{13} Hz

(B) 3×10^{14} Hz

(C) 3×10^{12} Hz

(D) 2×10^{12} Hz

Q56.

Which of the following represents the correct order of negative electron gain enthalpy for the elements O, S, Se and Te?

(A) Te < Se < S < O

(B) O < S < Se < Te

(C) O < S < Te < Se

(D) O < Te < Se < S

Q57.

An enzyme contains 0.56% of iron by mass in it. The number of Fe atoms in 1 g of this enzyme is (Atomic mass of Fe = 56 u)

(A) 0.1 N_A atoms

(B) 0.01 N_A atoms

(C) 0.0001 N_A atoms

(D) 0.05 N_A atoms

Q58.

Number of oxygen atoms in 224 ml of CO_2 at STP is

(A) 10^{-3} N_A

(B) 4×10^{-2} N_A

(C) 2×10^{-2} N_A

(D) 10^{-2} N_A

Q59.

Law of Multiple proportion is applicable to which pair of compounds?

(A) H_2S and H_2O

(B) HCl and HBr

(C) SO_2 and SO_3

(D) NaCl and KCl

Q60.

The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$. What is the atomic number of the element, which is just above the given element in the periodic table?

(A) 23

(B) 12

(C) 13

(D) 22

Q61.

For multielectronic species, the correct order of energy of the given orbitals is

(A) 3d > 4p > 5s > 5p

(B) 5p > 5s > 4p > 3d

(C) 4p > 3d > 5p > 5s

(D) 5s > 4p > 3d > 5p

Q62.

The pair of elements which have same value of electronegativity on Pauling scale, is

(A) Be and B

(B) C and S

(C) Al and P

(D) Na and Mg

Q63.

The correct order of first ionization enthalpy for the given elements is

(A) Be < B < C < N < O < F

(B) B < Be < C < O < N < F

(C) Be < B < C < O < N < F

(D) B < Be < C < N < O < F

Q64.

A 3.3 mg ball is moving with a speed of 200 m/s. The de Broglie wavelength associated with it will be ($h = 6.6 \times 10^{-34}$ Js)

(A) 10^{-27} m

(B) 10^{-30} m

(C) 10^{-24} m

(D) 10^{-33} m

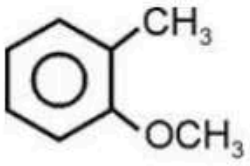
Q65.

Intramolecular hydrogen bond is present in

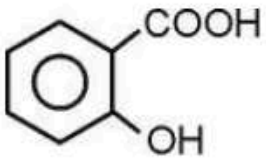
(A)



(B)



(C)



(D)



Q66.

Neutral oxide among the following is

(A) Cl_2O_7

(B) CO

(C) Al_2O_3

(D) Na_2O

Q67.

Given below are two statements one is labelled as assertion (A) other is labelled as reason (R).

Assertion (A): O_2 is a paramagnetic species.

Reason (R): O_2 has two unpaired electrons in π-antibonding molecular orbital.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Q68.

Total number of elements present in 3rd period of modern periodic table is

(A) 18

(B) 32

(C) 10

(D) 8

Q69.

An element X has the following isotopic composition ^{60}X : 60% and ^{62}X : 40%. The weighted average atomic mass of the naturally occurring

element X is

(A) 60.4 amu

(B) 60.5 amu

(C) 60.2 amu

(D) 60.8 amu

Q70.

Match atomic number in List-I with their IUPAC official name in List-II.

List-I	List-II
a. 101	(i) Bohrium
b. 103	(ii) Mendelevium
c. 105	(iii) Lawrencium
d. 107	(iv) Dubnium

The correct match is

(A) a(iii), b(ii), c(iv), d(i)

(B) a(ii), b(iii), c(i), d(iv)

(C) a(ii), b(iii), c(iv), d(i)

(D) a(ii), b(iv), c(iii), d(i)

Q71.

How many maximum electrons can be accommodated in an orbital for which n = 4 and l = 2?

(A) 5

(B) 10

(C) 2

(D) 6

Q72.

Match List-I with List-II.

List I (Electronic transition)		**List II (Maximum number of emission lines in H atom sample)**
a. $n_2 = 5$ to $n_1 = 1$	(i)	3
b. $n_2 = 3$ to $n_1 = 2$	(ii)	10
c. $n_2 = 4$ to $n_1 = 2$	(iii)	1
d. $n_2 = 6$ to $n_1 = 1$	(iv)	15

Choose the correct option.

(A) a(i), b(ii), c(iii), d(iv)

(B) a(iv), b(iii), c(ii), d(i)

(C) a(iii), b(i), c(iv), d(ii)

(D) a(ii), b(iii), c(i), d(iv)

Q73.

Consider the following two statements

Statement I: Number of electrons in bonding molecular orbital and antibonding molecular orbital are equal for Be_2.

Statement II: Bonding molecular orbital formed by two 1s atomic orbital has higher energy than corresponding antibonding molecular orbital.

In the light of above two statements choose the **correct** option.

(A) Statement I is incorrect but statement II is correct

(B) Statement I is correct but statement II is incorrect

(C) Both statement I and statement II are incorrect

(D) Both statement I and statement II are correct

Q74.

The molecule having see–saw shape is

(A) H_2O

(B) SF_4

(C) NH_3

(D) XeF_4

Q75.

Consider the following statements

(a) Shape of ozone molecule is bent.

(b) Ozone has 3 canonical forms.

(c) Formal charge on central oxygen atom in ozone is zero.

Choose the incorrect statement.

(A) (a) and (b) only

(B) (c) only

(C) (b) only

(D) (b) and (c) only

Q76.

Consider the following statements

Statement I : The total number of node(s) in 2*s* orbital and angular node(s) in 2*p* orbital are same.

Statement II : Total number of nodes in an orbital does not depend on the value of azimuthal quantum number.

In the light of above statements, choose the correct option

(A) Statement I is incorrect but statement II is correct

(B) Both statement I and statement II are incorrect

(C) Statement I is correct but statement II is incorrect

(D) Both statement I and statement II are correct

Q77.

The number of protons, neutrons and electrons in $^{175}_{71}\mathrm{Lu}$, respectively, are

(A) 104, 71 and 71

(B) 71, 71 and 104

(C) 175, 104 and 71

(D) 71, 104 and 71

Q78.

A 100 watt bulb emits monochromatic light of wavelength 300 nm. Calculate the number of photons emitted per second by the bulb

(A) 1.5×10^{20}

(B) 2×10^{20}

(C) 1.5×10^{19}

(D) 2×10^{19}

Q79.

The species which does not exist is

(A) H_2

(B) He_2

(C) B_2

(D) O_2

Q80.

All of the given species have equal number of σ and π bonds, except

(A) CO_2

(B) SO_2

(C) SO_3

(D) C_2H_2

Q81.

Molality of 20% (w/w) aqueous NaOH solution is

(A) 5.25 molal

(B) 1.25 molal

(C) 6.25 molal

(D) 4.75 molal

Q82.

Which among the following set of quantum numbers is incorrect?

(A) $n = 3,\ l = 2,\ m = 0,\ s = -\frac{1}{2}$

(B) $n = 3,\ l = 3,\ m = 0,\ s = +\frac{1}{2}$

(C) $n = 4,\ l = 3,\ m = +1,\ s = -\frac{1}{2}$

(D) $n = 4,\ l = 2,\ m = -2,\ s = -\frac{1}{2}$

Q83.

Match List-I with List-II

List-I Molecules	**List-II Total Number of non-bonding electron pairs**
(a) ClF_5	(i) 15
(b) $XeOF_4$	(ii) 13
(c) XeF_4	(iii) 16
(d) SF_4	(iv) 14

Choose the correct answer from the options given below

(A) (a) – (iii), (b) – (i), (c) – (ii), (d) – (iv)

(B) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

(C) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

(D) (a) – (iii), (b) – (i), (c) – (iv), (d) – (ii)

Q84.

The most covalent compound is

(A) LiI

(B) LiCl

(C) LiBr

(D) LiF

Q85.

Match List I with List II and choose the correct option.

List I (Energy level)	**List II (Energy in eV)**
(a) 2nd excited state of Li^{2+} ion	(i) $-13.6/4$
(b) 3rd excited state of He^{+} ion	(ii) -13.6
(c) Ground state of Be^{3+} ion	(iii) -13.6×16
(d) 3rd excited state of H atom	(iv) $-13.6/16$

(A) (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

(B) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Q86.

Consider the following statements

Statement I: Equal masses of S_2 and S_8 contain same number of molecules.

Statement II: Equal masses of O_2 and O_3 contain equal number of atoms.

In the light of above statements, choose the correct option.

(A) Both the statements I and II are correct

(B) Both the statements I and II are incorrect

(C) Statement I is correct but statement II is incorrect

(D) Statement I is incorrect but statement II is correct

Q87.

An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C = 40% and H = 6.67%. The empirical formula of the compound would be

(A) HCO

(B) H_2CO

(C) HC_2O

(D) H_2C_2O

Q88.

Correct order of dipole moment is

(A) $H_2O > HCl > NH_3$

(B) $H_2O > NH_3 > HCl$

(C) $NH_3 > H_2O > HCl$

(D) $HCl > H_2O > NH_3$

Q89.

The position of an element having atomic number 60 in modern periodic table is

(A) Sixth period, Second group

(B) Fifth period, Fifth group

(C) Sixth period, Third group

(D) Sixth period, Seventh group

Q90.

Choose the incorrect match

Base physical quantity	Name of SI unit
(1) Amount of substance	– Kilogram
(2) Length	– Metre
(3) Time	– Second
(4) Luminous intensity	– Candela

(A) (1)

(B) (2)

(C) (3)

(D) (4)

Q91.

Select the **incorrect** match w.r.t. various cells and their shape.

(A) Red blood cells – Round and biconcave

(B) White blood cells – Amoeboid

(C) Tracheid – Round and oval

(D) Columnar epithelial cells – Long and narrow

Q92.

Which of the following types of chromosomes have equal arms?

(A) Sub-metacentric chromosomes

(B) Telocentric chromosomes

(C) Metacentric chromosomes

(D) Acrocentric chromosomes

Q93.

The centriole in an animal cell

(A) Is made up of nine evenly spaced peripheral fibrils of tubulin protein

(B) Contains peripheral microtubule doublets interconnected by linkers

(C) Contains a pair of centrally located microtubules

(D) Shows 9 + 2 arrangement of microtubules

Q94.

Calcium pectate is chiefly present in the

(A) Primary cell wall

(B) Secondary cell wall

(C) Middle lamella

(D) Plasma membrane

Q95.

The Golgi apparatus

(A) Is found only in animals

(B) Is found in prokaryotes

(C) Is a site of rapid ATP production

(D) Modifies and packages proteins

Q96.

Eukaryotic flagella and cilia are similar to each other as both

a. Have '9+2' organisation of microtubules

b. Are surrounded by membrane

c. Help in locomotion

d. Give cartwheel appearance with no microtubules in the centre.

The **correct** one(s) is/are

(A) Only a and b

(B) Only c

(C) Only a and d

(D) Only a, b and c

Q97.

Select the ribosomal subunits found in bacteria.

(A) 60S and 40S

(B) 30S and 40S

(C) 50S and 30S

(D) 60S and 30S

Q98.

Which of the following statements is **incorrect** w.r.t. Endoplasmic Reticulum.

(A) RER is actively involved in protein synthesis and secretion

(B) SER is the major site for synthesis of lipids

(C) It divides the intracellular space into three distinct compartments

(D) In animal cells lipid like steroidal hormones are synthesised in SER

Q99.

Match List I with List II.

List I	List II
a. Nucleolus	(i) Infolding of mitochondrial inner membrane
b. Cristae	(ii) Site for ribosomal RNA synthesis

c. Cisternae	(iii)	Part of chromosome beyond secondary constriction
d. Satellite	(iv)	Flattened sac-like structures stacked on one another

Choose the **correct** answer from the following options.

(A) a(iii), b(ii), c(i), d(iv)

(B) a(ii), b(i), c(iv), d(iii)

(C) a(ii), b(i), c(iii), d(iv)

(D) a(iii), b(i), c(ii), d(iv)

Q100.

According to the cell theory

(A) All living cells have nucleus

(B) All cells arise from pre-existing cells

(C) Organisms are always composed of more than one cell

(D) Cell wall is a unique character of every cell

Q101.

Reserve material in prokaryotic cells are stored in the cytoplasm in the form of

(A) Inclusion bodies

(B) Nucleoid

(C) Plastids

(D) Ribosomes

Q102.

Read the given statements (A-D) and identify them as True(T) or False(F).

(A) Larger and more numerous nuclei are commonly present in cells carrying out protein synthesis.

(B) Chromatin contains DNA and some acidic proteins called histones, some non-histone proteins and also RNA.

(C) Nuclear pores are required for the exchange of RNA and proteins between nucleus and cytoplasm.

(D) Nuclear material stained by the basic dyes was given the name chromatin by Flemming.

Select the **correct** answer from the following options.

(A) A-(T), B-(F), C-(T), D-(F)

(B) A-(F), B-(F), C-(T), D-(T)

(C) A-(T), B-(F), C-(F), D-(T)

(D) A-(F), B-(T), C-(T), D-(F)

Q103.

Read the following events that happened in the field of cell biology in different years. Arrange the following in the chronological order of their occurrence and choose the **correct** option.

(A) An improved model of cell membrane structure was proposed by Singer and Nicolson

(B) Densely stained reticular structures near the nucleus were first observed by Camillo Golgi

(C) Nucleus as a cell organelle was first described by Robert Brown

(D) *Omnis cellula-e cellula* was explained by Rudolf Virchow

(A) $D \rightarrow C \rightarrow A \rightarrow B$

(B) $C \rightarrow A \rightarrow D \rightarrow B$

(C) $C \rightarrow D \rightarrow B \rightarrow A$

(D) $D \rightarrow A \rightarrow C \rightarrow B$

Q104.

The phase between two successive M phases is called

(A) Prophase

(B) Metaphase

(C) Anaphase

(D) Interphase

Q105.

The plant cytokinesis is different from animal cytokinesis, as the former

(A) Is achieved by the formation of a furrow in the cell membrane

(B) Shows wall formation which starts from the centre of the cell

(C) Shows organelle distribution between the two daughter cells at the time of cytoplasmic division

(D) Occurs by centripetal movement of cell wall

Q106.

In the S phase of cell cycle

(A) Chromosome number increases

(B) Cells become haploid

(C) Amount of DNA doubles

(D) Amount of DNA becomes half

Q107.

Which of the following events occurs once in meiosis?

(A) Recombination between homologous chromosomes

(B) Duplication of centrioles

(C) Karyokinesis

(D) Metaphasic plate formation

Q108.

Read the following events:

A. Disappearance of nuclear membrane.

B. Two groups of chromosomes get enclosed by separate nuclear membrane.

C. Shortening of microtubules attached to kinetochore aids the movement of sister chromatids towards opposite poles.

D. Microtubules from opposite poles of the spindle gets attached to the kinetochores of the sister chromatids.

Which of the following options represents the correct sequential order of the meiosis II events?

(A) $A \rightarrow D \rightarrow C \rightarrow B$

(B) $A \rightarrow D \rightarrow B \rightarrow C$

(C) $B \rightarrow A \rightarrow D \rightarrow C$

(D) $C \rightarrow B \rightarrow A \rightarrow D$

Q109.

How many of the following pairs is/are **correctly** matched?

a. Recombination nodule – Site at which crossing over occurs between sister chromatids

b. Prophase – Marked by complete disintegration of nuclear envelope

c. Zygotene – Sister chromatids in each chromosome are identical

d. Kinetochore – Junction between DNA and tubulin protein

e. Bivalent – Can be clearly seen as tetrad during metaphase

(A) Two

(B) Three

(C) Four

(D) One

Q110.

Read the statements given below and identify the **correct** option.

Statement A: All the cellular components are virtually reorganised during the most dramatic period of the cell cycle.

Statement B: Cells which do not divide further exit G_1 phase to enter an inactive stage where cell remain metabolically inactive and no longer proliferate unless called on to do so.

Choose the **correct** option.

(A) Both the statements A and B are correct

(B) Both the statements A and B are incorrect

(C) Statement A is correct but statement B is incorrect

(D) Statement B is correct but statement A is incorrect

Q111.

Enzyme recombinase activity is seen in

(A) Pachytene

(B) Zygotene

(C) Leptotene

(D) Diplotene

Q112.

In which stage of mitosis, chromosomes cluster at opposite spindle poles and their identity is lost as discrete element?

(A) Anaphase

(B) Telophase

(C) Metaphase

(D) Prophase

Q113.

Read the following statements and select the **correct** option.

Statement A: Cell cycle is the period between the formation of new daughter cells and their further division.

Statement B: Interphase is a biosynthetic phase in which cell duplicates its organelles and replicates its DNA.

(A) Both statements A and B are correct

(B) Both statements A and B are incorrect

(C) Only statement A is correct

(D) Only statement B is correct

Q114.

If there are 40 chromosomes in G_1-phase, what will be the number of bivalents in zygotene stage?

(A) 30

(B) 80

(C) 10

(D) 20

Q115.

State true (T) or false (F) for the given statements and choose the **correct** option.

A. During G_2 phase, proteins are synthesised in preparation for mitosis while cell growth continues.

B. The plants and animals can show mitotic division in both haploid and diplod cells.

	A	B
(1)	T	T
(2)	F	T
(3)	F	F
(4)	T	F

(A) (1)

(B) (2)

(C) (3)

(D) (4)

Q116.

ICBN assigns rules for the scientific naming of

(A) Bacteria

(B) Plants

(C) Viruses

(D) Insects

Q117.

Observe the following taxa associated with wheat:

a. Poales

b. Poaceae

c. Angiospermae

d. Plantae

e. Monocotyledonae

Which of the following options **correctly** depicts the sequential order of these taxa in ascending order?

(A) a → d → b → e → c

(B) b → a → e → c → d

(C) b → e → c → a → d

(D) c → a → d → b → e

Q118.

Datura is placed in the family

(A) Poaceae

(B) Anacardiaceae

(C) Solanaceae

(D) Muscidae

Q119.

Select the **incorrect** match w.r.t. taxonomic categories of man.

(A) Species – *sapiens*

(B) Order – Primata

(C) Class – Chordata

(D) Kingdom – Animalia

Q120.

Out of the seven taxonomic categories, the basic and the lowest category of taxonomic hierarchy is

(A) Species

(B) Genus

(C) Tribe

(D) Family

Q121.

Rules of Binomial nomenclature were given by

(A) Robert Hooke

(B) Rudolf Virchow

(C) Carolus Linnaeus

(D) Ernst Mayr

Q122.

Select the **incorrect** statement.

(A) The branch of science dealing with classification, nomenclature, characterisation and identification of organisms is referred to as taxonomy

(B) The members within the order Polymoniales share more common characteristics than that of the family Convolvulaceae.

(C) The problem of classification becomes more complex as we move from lower to higher taxa.

(D) The earliest classifications were based on the 'uses' of various organisms.

Q123.

Potato and mango belong to the same

(A) Family

(B) Genus

(C) Phylum

(D) Class

Q124.

Bacteriophages usually have

(A) dsDNA

(B) ssDNA

(C) dsRNA

(D) ssRNA

Q125.

In five kingdom classification, only single celled eukaryotes are included in

(A) Monera

(B) Protista

(C) Fungi

(D) Plantae

Q126.

Dikaryophase is **not** seen in

(A) Ascomycetes

(B) Phycomycetes

(C) Basidiomycetes

(D) Both Ascomycetes and Basidiomycetes

Q127.

Which of the given protozoans causes the disease sleeping sickness?

(A) *Paramoecium*

(B) *Trypanosoma*

(C) *Plasmodium*

(D) *Entamoeba*

Q128.

Read the given statements.

(a) Found in terrestrial habitats and on decaying wood in moist and damp places.

(b) The hyphae are continuous tubes filled with multinucleated cytoplasm.

(c) The asexual spores are generally conidia produced on conidiophores.

(d) A transitional phase called dikaryophase appears in the life cycle.

(e) The gametes can be isogamous, anisogamous or oogamous.

Which of the above characteristics is/are **true** for the class to which *Rhizopus* belongs?

(A) (a), (b) and (c) only

(B) (b), (c) and (d) only

(C) (a), (b) and (e) only

(D) (c), (d) and (e) only

Q129.

The common field mushroom with edible basidiocarp is

(A) *Puccinia*

(B) *Claviceps*

(C) *Agaricus*

(D) *Aspergillus*

Q130.

Mark the **incorrect** option w.r.t. bacteria

(A) Bacterial cell structure is very complex but it has simple behaviour

(B) They show most extensive metabolic diversity

(C) They may be photosynthetic autotrophic or chemosynthetic autotrophic

(D) Ribosomes are 70S type

Q131.

Mark **incorrect** statement w.rt. *Nostoc* .

(A) It is photosynthetic

(B) It fixes atmospheric nitrogen

(C) It has heterocyst

(D) It is flagellated unicellular form

Q132.

Identify the **correct** statements w.r.t. infectious agent discovered by T.O. Diener

A. It is larger than virus

B. Contains low molecular weight RNA

C. Causes potato spindle tuber disease

D. Has a protein coat called capsid

(A) A and D

(B) B and C

(C) B and D

(D) A and C

Q133.

Read the following statements and select the **correct** option.

(A)

Paramoecium and *Amoeba* were placed in the kingdom Protista in the earliest systems of classification.

(B)

Members of the kingdom Fungi show heterotrophic mode of nutrition.

(C)

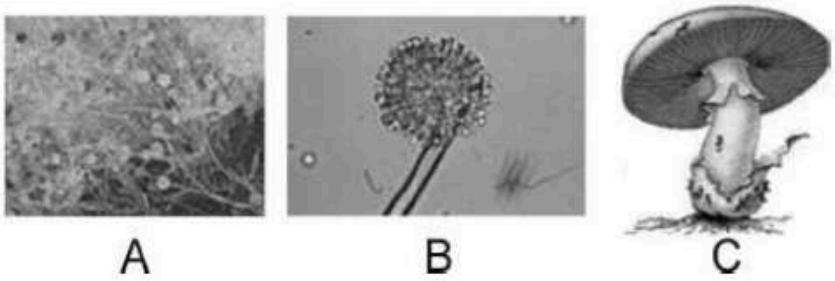
Five kingdom classification, given by Whittaker, did not consider phylogenetic relationships.

(D)

Three-domain system of classification divided all the eukaryotic organisms in two domains.

Q134.

Identify A, B and C in given diagram and choose the **correct** option.



(A)

A = *Mucor*, B = *Aspergillus*, C = *Agaricus*

(B)

A = *Mucor*, B = *Agaricus*, C = *Aspergillus*

(C)

A = *Agaricus*, B = *Mucor*, C = *Aspergillus*

(D)

A = *Agaricus*, B = *Aspergillus*, C = *Mucor*

Q135.

Read the given statements (a-d).

a. Methanogens are present in the gut of several ruminant animals.

b. Chemosynthetic autotrophic bacteria oxidise nitrates, nitrites and ammonia and use the released energy for their ATP production.

c. Cyanobacteria often form blooms in polluted water bodies.

d. Thermoacidophiles are found in extreme salty areas and cannot survive in water having low pH.

In the light of above statements, select the **correct** option.

(A) All are correct except d

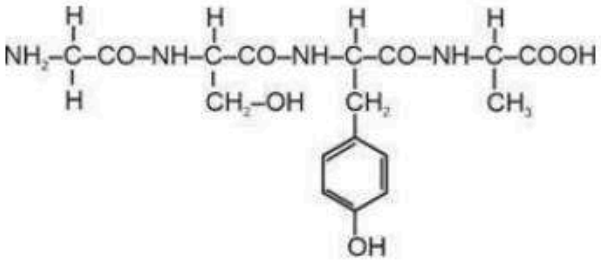
(B) Only a and b are correct

(C) Only c and d are correct

(D) Only b and c are correct

Q136.

Choose the correct option which provides the suitable names to the amino acids involved in the formation of the given structure of an oligopeptide starting from first amino acid



(A) Glycine–Alanine–Serine–Tyrosine

(B) Glycine–Serine–Alanine–Tyrosine

(C) Alanine-Serine-Tyrosine-Glycine

(D) Glycine-Serine-Tyrosine-Alanine

Q137.

Choose the **incorrect** statement.

(A) Filtrate and retentate are two fractions obtained during chemical analysis of a living tissue.

(B) Inorganic compounds like sulphates and phosphates are found in acid-insoluble fraction.

(C) All the organic compounds that we get from living tissues can be called biomolecules.

(D) Oxygen constitutes the maximum percentage weight of Earth's crust as well as the human body.

Q138.

The most abundant protein present in the animal world is ‘P’.

Select the **correct** option for ‘P’.

(A) Also found in the blood

(B) Arranged in an irregular pattern in dense connective tissue that is present beneath the skin

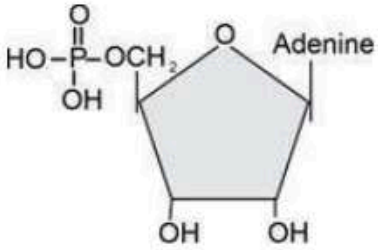
(C) Does not contain the hydrogen bonds

(D) A storage homopolymer in plants

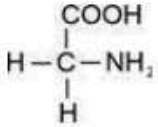
Q139.

Among the biomolecules depicted below, choose the one that will be found in the retentate portion obtained upon the chemical analysis of a living tissue.

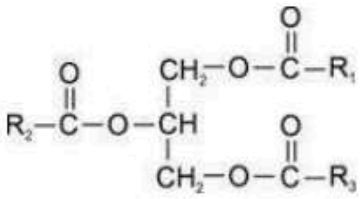
(A)



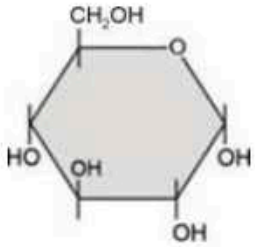
(B)



(C)



(D)



Q140.

Total number of carbon atoms present in arachidonic acid and palmitic acid respectively are

(A) 16 and 20

(B) 18 and 20

(C) 20 and 16

(D) 22 and 16

Q141.

Dry weight of a living tissue differs from its wet weight as

(A) Latter is devoid of water

(B) Former is devoid of water

(C) Latter is devoid of inorganic salts

(D) Former is devoid of water, CO_2 and inorganic salts

Q142.

Read the items listed in box below.

Collagen, RuBisCO, Chitin, Inulin, Cellulose, Glycogen

How many among them is/are structural homopolymers?

(A) One

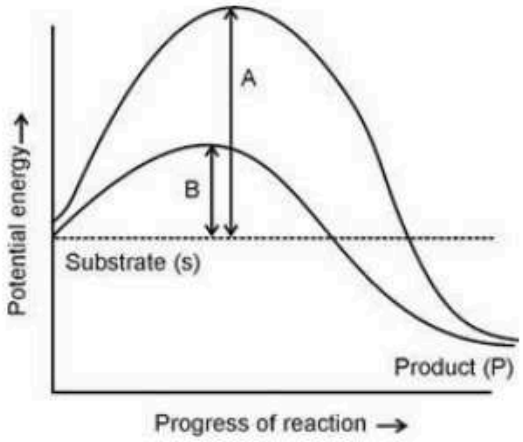
(B) Three

(C) Two

(D) Four

Q143.

Which of the following is the **correct** description w.r.t. given graph of a reaction catalysed with and without enzyme?



(A) A : Activation energy of an endothermic reaction without enzyme

(B) B : Activation energy of an exothermic reaction with enzyme

(C) A : Activation energy of an exothermic reaction with enzyme

(D) B : Activation energy of an endothermic reaction without enzyme

Q144.

A student has samples of certain compounds which are listed below:

A – Collagen

B – Glycogen

C – Palmitic acid

D – DNA

E – Calcium ions

His mentor asked him to arrange the category to which these compounds belong to in the increasing order w.r.t. their per cent in the total cellular mass. Which among the following will be the correct sequence?

(A) B < A < C < D < E

(B) E < C < B < D < A

(C) A < B < D < C < E

(D) E < A < C < D < B

Q145.

Match column I with column II w.r.t. given compounds.

Column I	Column II
a. Adenine	(i) 1 heterocyclic ring
b. Uracil	(ii) 4 homocyclic rings
c. Cholesterol	(iii) 2 heterocyclic rings
d. Adenylic acid	(iv) 3 heterocyclic rings

Select the **correct** option.

(A) a(iii), b(i), c(ii), d(iv)

(B) a(i), b(iv), c(iii), d(ii)

(C) a(iv), b(iii), c(i), d(ii)

(D) a(ii), b(iii), c(iv), d(i)

Q146.

'X' is the prosthetic group of an enzyme named catalase. 'X' is also the prosthetic group in

(A) Carboxypeptidase

(B) Peroxidase

(C) Carbonic anhydrase

(D) Succinic dehydrogenase

Q147.

Assertion (A): Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition.

Reason (R): Malonate closely resembles the substrate of succinic dehydrogenase in the structure.

In the light of the above statements, choose the correct option.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q148.

Assertion (A): The valve which guards the opening between right atrium and right ventricle of human heart is called bicuspid valve.

Reason (R): The valve that guards the opening between right atrium and right ventricle is formed by two muscular cusps.

In the light of above statemetns, choose the correct answer from the options given below.

(A) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is true but (R) is false

(D) Both (A) and (R) are false

Q149.

Which of the following carbohydrates can be broken into simpler units *via* the action of the enzymes that have the EC number 3.2.1.20?

(A) Ribose

(B) Glucose

(C) Galactose

(D) Maltose

Q150.

Aggregates of cell membranes as vesicles after homogenisation are obtained in which fraction after chemical analysis of a living animal tissue?

(A) Acid-soluble fraction

(B) Acid-insoluble fraction

(C) Filtrate fraction

(D) Micromolecular fraction

Q151.

Neurons contain

(A) Myosin

(B) Sacroplasm

(C) Sarcolemma

(D) Dendrites

Q152.

The function of adhering junctions is to

(A) Facilitate communication between adjoining cells by connecting the cytoplasm for rapid transfer of ions, small molecules and some large molecules.

(B) Separate two cells from each other.

(C) Perform cementing to keep neighbouring cells together.

(D) Stop substances from leaking across a tissue.

Q153.

All of the following are the characteristic features of epithelium forming lining of alveoli, **except**

(A) A single thin layer of flattened cells with irregular boundaries

(B) Cells are compactly packed with little intercellular matrix

(C) The cells are ciliated

(D) The nuclei of cells are flat and often lie at the centre of cells

Q154.

Which of the following given features are **true** w.r.t cardiac muscle fibres?

(a) They are branched

(b) They do not get fatigued

(c) Nuclei are peripheral in position

(d) Non striated

(e) Cylindrical in shape

Select the **correct** option.

(A) (a), (b) and (e)

(B) (a), (b), (c) and (e)

(C) (a), (b), (d) and (e)

(D) (a), (b) and (d)

Q155.

The excess of nutrients which are not used immediately in the body are converted into fats and are mainly stored in _____.

Choose the option which fills the blank **correctly**.

(A) Areolar connective tissue

(B) Adipose connective tissue

(C) Dense regular connective tissue

(D) Dense irregular connective tissue

Q156.

Multi-layered epithelium that has limited role in secretion and absorption but provides protection against chemical and mechanical stresses is **not** found in

(A) Dry surface of skin

(B) Moist surface of pharynx

(C) Inner lining of pancreatic duct

(D) Inner lining of oviduct

Q157.

'X' is a specialised connective tissue. It is the site of production of cells of another specialised connective tissue 'Y'. Identify 'X' and 'Y' and select the **correct** option.

(A) Matrix of 'Y' is solid, hard and non-pliable

(B) Matrix of 'X' is hard and pliable

(C) Skeletal muscles are found attached to 'X' to bring about movements

(D) 'X' and 'Y' both contain the semi-fluid ground substance and fibres

Q158.

Cells of epithelial lining of stomach contain

(A) Gap junctions to block transfer of ions between adjacent cells

(B) Intercalated discs for keeping cells together

(C) Tight junctions to inhibit the passage of substances between cells

(D) Adhering junctions for communication with glands

Q159.

Division of labour due to tissue formation for first time became evident in the members of the phylum

(A) Porifera

(B) Platyhelminthes

(C) Echinodermata

(D) Cnidaria

Q160.

Areolar tissue differs from adipose tissue as it

(A) Actively participates in inflammation and repair due to presence of mast cells

(B) Is a type of connective tissue that contains elastic and collagen fibres scattered randomly in its matrix.

(C) Is the abundantly distributed loose connective tissue and fills up the space between organs contrary to adipose tissue which primarily acts as a thermal insulator.

(D) Does not possess fat storing cells

Q161.

The enzyme carbonic anhydrase

(A) Belongs to the class I of enzymes

(B) Is found in the most abundant formed element of blood that contains the respiratory pigment

(C) Is not present in the blood plasma

(D) Contains Ca^{+2} as its prosthetic group

Q162.

At the level of which vertebra does trachea divide into right and left primary bronchi?

(A) 4th thoracic vertebra

(B) 5th lumbar vertebra

(C) 5th thoracic vertebra

(D) 5th cervical vertebra

Q163.

If a healthy adult man inspires or expires 9000 mL of air per minute, then what will be his breathing rate per minute assuming his tidal volume remains normal?

(A) 14 times

(B) 16 times

(C) 18 times

(D) 19 times

Q164.

The oxygen – haemoglobin dissociation curve will show a left shift in case of

(A) High temperature

(B) High partial pressure of carbon dioxide

(C) High partial pressure of oxygen

(D) High hydrogen ion concentration

Q165.

Peripheral chemoreceptors that help in regulation of breathing are present in

(a) Aortic arch

(b) Carotid artery

(c) Medulla oblongata

Choose the **correct** option.

(A) (a) only

(B) (b) only

(C) (a) and (b) only

(D) (a) and (c) only

Q166.

Read the given statements and select the **correct** option.

Statement A: Normal expiration is a passive process in humans and no muscular contractions are involved.

Statement B: Normal expiration results from the elastic recoil of the lungs.

(A) Only statement A is correct

(B) Only statement B is correct

(C) Both statements A and B are correct

(D) Both statements A and B are incorrect

Q167.

Structures which are **not** supported by incomplete cartilaginous rings are

(A) Trachea

(B) Primary bronchi

(C) Initial bronchi

(D) Terminal bronchioles

Q168.

Lungs are present in thoracic chamber which is ventrally, dorsally and laterally formed respectively by

(A) Vertebral column, ribs, sternum

(B) Ribs, sternum, diaphragm

(C) Sternum, vertebral column, ribs

(D) Diaphragm, sternum, ribs

Q169.

Select the **correct** match between respiratory volume/capacity and their value w.r.t. a normal adult human.

(A) Residual volume : 500 mL

(B) Vital capacity : 1000-1100 mL

(C) Inspiratory capacity : 2300-7000 mL

(D) Functional residual capacity : 2100-2300 mL

Q170.

Complete the analogy by selecting the **correct** option w.r.t. respiratory structures.

Moist cuticle : Earthworm : : Lung : _____

(A) Aquatic insects

(B) Aquatic mammals

(C) Fishes

(D) Aquatic molluscs

Q171.

Complete the analogy w.r.t. disorders of human circulatory system.

Angina pectoris : Acute chest pain :: Heart failure : _____

Choose the **correct** option.

(A) Atherosclerosis

(B) Sudden damage of heart muscle

(C) Reduced cardiac output

(D) Stoppage of the heart beat

Q172.

In a standard ECG, the T-wave represents

(A) Depolarisation of atria

(B) Repolarisation of ventricles

(C) Depolarisation of ventricles

(D) Electrical excitation of atria

Q173.

Blood from intestine is carried to the liver via _____ before it is returned to systemic circulation. The **correct** option which fills the blank is

(A) Hepatic vein

(B) Hepatic portal vein

(C) Hepatic artery

(D) Inferior mesenteric artery

Q174.

Match column I with column II.

Column I		**Column II**
(a) Fishes	(i)	Both pulmonary and systemic circulations are present
(b) Amphibians	(ii)	Most of them contain 3-chambered heart
(c) Reptiles	(iii)	Both oxygenated and deoxygenated blood mix in a single ventricle

(d) Birds (iv) Heart pumps out deoxygenated blood

Choose the **correct** option.

(A) a(iv), b(iii), c(i), d(ii)

(B) a(iv), b(ii), c(i), d(iii)

(C) a(iv), b(iii), c(ii), d(i)

(D) a(i), b(ii), c(iii), d(iv)

Q175.

Neutrophils and monocytes differ from each other in all of the following **except**

(A) The shape of their nuclei in mature form

(B) Their contribution to total WBC content of blood

(C) Their mode of action through phagocytosis against foreign organisms

(D) Their cytoplasmic contents

Q176.

Each haemoglobin molecule has

(A) One heme group

(B) Two heme groups

(C) Three heme groups

(D) Four heme groups

Q177.

Read the given statements

Statement A : Individuals with blood group AB are called universal acceptors.

Statement B : Individuals with AB blood group harbour anti-A, B antibodies in their plasma.

Choose the **correct** option.

(A) Both statements A & B are incorrect

(B) Statement B explains statement A

(C) Statement B is correct but does not explain statement A

(D) Statement A is correct

Q178.

A person 'X' met with an accident and immediately requires blood transfusion. On blood examination, it is observed that his blood contains anti-A and anti-B antibodies in plasma and no blood group antigens on RBCs. The donor's blood group most suitable for transfusion in 'X' is

(A) A

(B) B

(C) AB

(D) O

Q179.

An athlete has more cardiac output than normal due to

a. Change in stroke volume

b. Change in heart rate

c. Parasympathetic stimulation

d. More release of acetylcholine near SA node

Choose the **correct** option.

(A) a & b

(B) a, b & c

(C) a, b, c & d

(D) b only

Q180.

A Rh –ve mother gave birth to her first baby who is Rh +ve. To prevent hemolytic disease in her second foetus, she should be immediately administered with

(A) Rh antigen

(B) Anti-Rh antibodies

(C) Bone marrow

(D) Antibiotics